Hans Gao

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EDUCATION

San Jose State University (SJSU), San Jose, CA *Expecting Spring 2028*
B.S. Computer Science

De Anza College, Cupertino, CA *September 2024 – June 2026*
Computer Science Transfer | Relevant Coursework: Algorithms & Data Structures, Advanced C++ Programming

EXPERIENCE

Robotics Club — Operations Lead *February – June 2026*
Tech Stack: OnShape

- Designed the weapon system for a 1-lb combat robot in OnShape
- Coordinated logistics for a planned multi-college battle bot competition (Stanford, De Anza, Foothill, Mission), managing part orders, delivery timing, and inventory across all participating teams

De Anza Aerospace Club

President *November 2025 – June 2026*

- Grew membership from 4 to 15+ and led OnShape training sessions to prepare the team for the Spring 2026 two-stage rocket build
- Initiated a partnership with De Anza's manufacturing department, launching joint laser cutting workshops

Tech Lead *September 2025 – June 2026*
Tech Stack: OnShape, OpenRocket, Arduino IDE (C++)

- Directed CAD design of the two-stage rocket, nosecone, and avionics bay in OnShape; used OpenRocket to model apogee and tune fin design for stability
- Led the electronics team in designing and programming a custom flight computer
- Proposed and led early-stage research on a Kalman filter with a 5-person team to optimize second-stage ignition timing and parachute deployment; evaluated existing EKF libraries before deciding to build a custom implementation

LEADERSHIP & ACTIVITIES

STEM Zone 2026 — De Anza Representative *April 2026*

- Presented and tabled for De Anza's Aerospace Club, Robotics Club, and Special Engineering Projects course to prospective students
- Demonstrated a robotic arm built by the Special Projects team; modified its Python control script on the spot to run a cyclical motion pattern, reducing manual operation

PROJECTS

Personal Website (https://pit-tac.github.io/personal-website/) *June 2026 – Present*
Tech Stack: Cursor, HTML, CSS, JavaScript

- Built a personal site to showcase projects
- Used JavaScript to make interactive scroll elements and animations

Durak Game AI Agent (https://github.com/Pit-tac/Durak) *November 2025*
Tech Stack: Claude Code, Godot, GDScript, Git

- Built a Godot game based on Durak, a Russian trick-taking card game
- Developed a custom Monte Carlo Tree Search algorithm with expectimax for the game AI

Top-down Shooter Game *2024 – 2025*
Tech Stack: Godot, GDScript, Git

- Designed a top-down shooter game in Godot
- Implemented basic physics and swarm simulation for enemy movement

Pseudorandom Hash *December 2024*
Tech Stack: C++

- Programmed a pseudorandom hash table with quadratic probing for hash collision mitigation in C++
- Fixed unresolved memory leak issues from existing code base

Binary Search ADT *October 2024*
Tech Stack: C++

- Built a binary search algorithm in C++
- Expanded on previous work on queues, linked-lists, and overloaded comparison operators

SKILLS

Languages: C++, GDScript, HTML/CSS, JavaScript, Python
Tools: Claude Code, Excel, Cursor, Github Copilot, Git, Godot, Microsoft Office, OnShape, Outlook, OpenRocket, Arduino IDE
Concepts: Data Structures & Algorithms, Kalman Filtering, Monte Carlo Tree Search, Expectimax, CAD & Simulation